

Description of Additional Supplementary Files

Supplementary Data 1: Effect size of the PRS for the seven screening diseases in the UK Biobank testing set, across genetically inferred ancestry groups where case count exceeds 100.

Ln Odds Ratio = Log of the odds ratio per 1 standard deviation of the PRS; Odds Ratio = odds ratio per 1 standard deviation of the PRS.

LCI = lower confidence interval; UCI = upper confidence interval.

Estimates are obtained from logistic regression with age and sex as covariates, using both sexes together except for breast cancer (BC) and prostate cancer (PC).

AAA = abdominal aortic aneurysm; BC = female breast cancer; CRC = colorectal cancer; CAD = coronary artery disease; HT = hypertension; PC = prostate cancer; T2D = type 2 diabetes.

AFR = Sub-Saharan African ancestry; EAS = East Asian ancestry; EUR = European ancestry; SAS = South Asian ancestry.

Supplementary Data 2: Estimated risk advancement period (RAP) in years for the seven screening diseases for those at high ($OR > 2$), very high ($OR > 3$) and low ($OR < 0.5$) PRS.

AAA = abdominal aortic aneurysm; BC = female breast cancer; CRC = colorectal cancer; CAD = coronary artery disease; HT = hypertension; PC = prostate cancer; T2D = type 2 diabetes.

Supplementary Data 3: Estimates of clinical benefits of PRS-guided screening, exploring alternative conventional starting ages for screening in breast cancer, colorectal cancer and coronary artery disease.

Numbers in the bottom three rows report expected numbers of cases within the RAP in the high PRS risk group, expected deaths from those cases without screening, and expected deaths prevented by PRS guided screening, per 100k population of the corresponding sex.

Numbers in brackets are 95% confidence intervals. AAA = abdominal aortic aneurysm; BC = female breast cancer; CRC = colorectal cancer; CAD = coronary artery disease; HT = hypertension; PC = prostate cancer; T2D = type 2 diabetes.

Supplementary Data 4: Estimates of clinical benefits of PRS-guided screening leading to early detection benefit for very high PRS risk groups ($OR > 3$).

Conventional screening ages of 50 and 45 years have additionally been explored for breast cancer and colorectal cancer, respectively.

Numbers in the bottom three rows report expected numbers of cases within the RAP in the very high PRS risk group, expected deaths from those cases without screening, and expected deaths prevented by PRS guided screening, per 100k population of the corresponding sex.

Numbers in brackets are 95% confidence intervals. AAA = abdominal aortic aneurysm; BC = female breast cancer; CRC = colorectal cancer; CAD = coronary artery disease; HT = hypertension; PC = prostate cancer; T2D = type 2 diabetes.

Supplementary Data 5: Estimates of clinical benefits of PRS-guided screening leading to early prevention benefit for very high PRS risk groups ($OR > 3$).

Numbers in the bottom three rows report expected numbers of cases within the RAP in the very high PRS risk group, expected deaths from those cases without screening, and expected deaths prevented by PRS guided screening, per 100k population of the corresponding sex.

Numbers in brackets are 95% confidence intervals. AAA = abdominal aortic aneurysm; BC = female breast cancer; CRC = colorectal cancer; CAD = coronary artery disease; HT = hypertension; PC = prostate cancer; T2D = type 2 diabetes.

Supplementary Data 6: Empirical clinical benefits of PRS-guided screening leading to early detection or prevention benefit for high PRS risk groups ($OR > 2$), using cases observed in the UK Biobank testing set.

Alternative conventional starting ages for screening are also provided.

Due to limited sample size, values for non-sex-specific diseases are combined across both sexes.

Values for AAA and PC are for males only. Values for BC are for females only.

Numbers in the bottom three rows report expected numbers of cases within the RAP in the high PRS risk group, expected deaths from those cases without screening, and expected deaths prevented by PRS guided screening, per 100k population of the corresponding sex (for AAA, BC and PC), or per 100k of whole population (for other diseases).

Numbers in brackets are 95% confidence intervals. AAA = abdominal aortic aneurysm; BC = female breast cancer; CRC = colorectal cancer; CAD = coronary artery disease; HT = hypertension; PC = prostate cancer; T2D = type 2 diabetes.

Supplementary Data 7: Empirical clinical benefits of PRS-guided screening leading to early detection or prevention benefit for very high PRS risk groups ($OR > 3$), using cases observed in the UK Biobank testing set.

Alternative conventional starting ages for screening are also provided.

Due to limited sample size, values for non-sex-specific diseases are combined across both sexes.

Values for AAA and PC are for males only. Values for BC are for females only.

Numbers in the bottom three rows report expected numbers of cases within the RAP in the high PRS risk group, expected deaths from those cases without screening, and expected deaths prevented by PRS guided screening, per 100k population of the corresponding sex (for AAA, BC and PC), or per 100k of whole population (for other diseases).

Numbers in brackets are 95% confidence intervals. AAA = abdominal aortic aneurysm; BC = female breast cancer; CRC = colorectal cancer; CAD = coronary artery disease; HT = hypertension; PC = prostate cancer; T2D = type 2 diabetes.

Supplementary Data 8: Estimates of clinical benefits of PRS-guided screening in high PRS risk groups ($OR > 2$), using UK-based national sources of disease incidence and mortality.

Numbers in the bottom three rows report expected numbers of cases within the RAP in the high PRS risk group, expected deaths from those cases without screening, and expected deaths prevented by PRS guided screening, per 100k population of the corresponding sex.

Numbers in brackets are 95% confidence intervals. AAA = abdominal aortic aneurysm; BC = female breast cancer; CRC = colorectal cancer; CAD = coronary artery disease; HT = hypertension; PC = prostate cancer; T2D = type 2 diabetes.